Bingchen Wang

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Research Profile

My research develops a theory and practice of intelligence under constraints: how human, artificial, and hybrid systems learn, coordinate, and align under informational, economic, and institutional limits. Across projects, I build methods that reduce the fixed cost of asking serious empirical questions about AI systems, relaxing dependence on abundant compute, sensitive data, proprietary access, or bespoke human pipelines. Recent work spans preference alignment without demographic data or finetuning, incentive design for collaborative machine learning, synthetic-agent feedback for product iteration, and auditing of algorithmic bias.

Education

University of Oxford, St Hilda's College — Oxford, UK
M.Phil. in Economics *(Distinction)* — Oct 2019 – Jun 2021
Thesis: *Detecting Parameter Shifts using Multiplicative-Indicator Saturation*.
Supervisors: David Hendry and Jennifer Castle

Columbia University — New York, USA
B.A. in Mathematics–Statistics *(Summa cum laude)* — Sep 2016 – Dec 2018
Phi Beta Kappa; Department Honors in Statistics; GPA: 4.13/4.33

City University of Hong Kong — Hong Kong SAR
B.Sc. (Hons.) in Computing Mathematics *(First Class Honours)* — Sep 2014 – May 2016
Joint Bachelor's Degree Program with Columbia University; GPA: 4.13/4.3

Publications and Working Papers

Bingchen Wang, Zi-Yu Khoo, Jingtan Wang. "Prompts to Proxies: Emulating Human Preferences via a Compact LLM Ensemble." *arXiv preprint* arXiv:2509.11311, 2026. Under review

Bingchen Wang, Zhaoxuan Wu, Fusheng Liu, Bryan Kian Hsiang Low. "Paid with Models: Optimal Contract Design for Collaborative Machine Learning." *Proceedings of the AAAI Conference on Artificial Intelligence (AAAI-25)*, Oral Presentation, Philadelphia, USA, 2025

Research Systems and Software

P2P — Preference reconstruction with compact LLM proxy ensembles — *Pre-release (site)*
Python; LLM APIs; survey-response pipelines; constrained Lasso/ElasticNet; held-out validation

- Provides an inference-time framework for reconstructing population-level preferences from LLM proxy agents without finetuning, sensitive demographic conditioning, or individual-level digital twins.
- Supports structured proxy-agent generation, survey-response collection, response cleaning, entropy-guided expansion, sparse weighted aggregation, held-out validation, and automated reporting.
- Evaluates reconstruction fidelity using held-out MSE, entropy, distributional diagnostics, and response-diversity measures across empirical survey benchmarks.

SIM-PANEL — Synthetic panel simulation and benchmarking toolkit — *Open source (GitHub)*
Python; CLI/YAML; JSON/JSONL artifacts; schema validation; streaming ingestion; distributional diagnostics

- Provides a CLI-driven, YAML-configured toolkit for generating, validating, and benchmarking synthetic panel-style datasets for LLM-agent simulation research.
- Supports randomized assignment, manual assignment, and agent self-selection policies for studying behavioral responses under different treatment-allocation and selection assumptions.
- Includes source adapters, benchmark builders, comparison workflows, schema validation, JSON/JSONL artifacts, automated reports, and distributional diagnostics against real or controlled reference data.

Research Experience

Independent Researcher Remote / Zibo, China
Independent Research — AI, Economics and Computational Social Science Jul 2025 – Present

- Originated and completed *Prompts to Proxies*, a preference-reconstruction framework that challenges digital-twin-style population simulation by using LLM proxy ensembles to reconstruct aggregate survey-response distributions under weak information and compute constraints.
- Built the `P2P` research system end to end, from proxy generation and survey-response collection to sparse aggregation, held-out validation, and automated reporting.
- Developed `SIM-PANEL`, a synthetic-panel simulation and benchmarking toolkit for evaluating LLM-agent behavioral responses under randomized assignment, manual assignment, and self-selection designs.
- Currently working on an econometric sequel to *Prompts to Proxies* and the design of a controlled audit framework for evaluating bias mechanisms in LLM-mediated hiring and résumé assessment.
- Manage independent research workflows spanning theory, implementation, cloud/API experimentation, empirical validation, documentation, paper writing, and public-facing communication.

Research Assistant Singapore
Institute of Data Science, National University of Singapore Jul 2023 – Jun 2025

- Originated and completed *Paid with Models*, a contract-theoretic framework for incentivizing resource contribution in collaborative machine learning using models as rewards.
- Formulated the principal-agent problem under private contribution costs, derived theoretical properties of optimal contracts, and designed numerical experiments to study welfare implications.
- Prepared the manuscript, figures, rebuttal, poster, oral presentation, and invited-talk materials; paper accepted as an AAAI-25 oral presentation.

Independent Learning and Research Hong Kong & Zibo, China
Transition from Social Sciences to AI May 2022 – Jun 2023

- Completed structured coursework in *Python*, *Machine Learning*, and *Deep Learning*; studied core texts such as *The Elements of Statistical Learning*; explored literature on federated learning.
- Conducted exploratory projects on text analysis, clustering, and policy simulation using Python and PyTorch; authored learning materials and open notebooks as *Learner's Corner*.

Senior Research Assistant Hong Kong SAR
Research Hub on Institutions of China, The University of Hong Kong Sep 2021 – Apr 2022

- Cleaned, linked, and analyzed large-scale Chinese survey datasets (CGSS, CLDS, CFPS); conducted exploratory data analysis and data-quality checks in R.
- Applied ordered logistic regression and grouped Lasso models to study associations among ideology, perceived income justice, and family background.
- Drafted empirical-analysis sections and supported manuscript evaluation through reviews of research design, data quality, econometric methods, and robustness checks.

Short-Term Consultant Remote / Washington, DC, USA
The World Bank Sep 2021

- Supported fiscal and economic analysis for the Jordan MTI team by updating debt/fiscal databases, oil-price monitors, Economic Monitor workflows, and Debt Sustainability Analysis materials.
- Automated recurring data updates in R, reducing processing time from over 30 minutes to under 5 minutes while improving reproducibility and reliability.

Research Assistant Oxford, UK
Climate Econometrics Group, University of Oxford Oct 2019 – Jun 2021

- Maintained and expanded large-scale datasets on anthropogenic CO_2 emissions, integrating data from

NASA, FRED, OECD, and national sources; produced documentation and visualizations supporting reproducible dataset updates and empirical analysis.
- Conducted econometric analysis on parameter-shift detection and time-series diagnostics; work contributed to MPhil thesis on multiplicative-indicator saturation (MIS).

Presentations

Invited Talk: "Let's Talk About Trust — Paid with Models," Trusted CollabML Lab End-of-Project Meeting, Singapore, Mar 2025.

Workshop Presentation: "Fine-Tuning LLMs with Noisy Data for Political Argument Generation," GoodData Workshop, AAAI-25, Philadelphia, USA, Mar 2025. Presented on behalf of S. Churina and K. Jaidka.

Oral Presentation: "Paid with Models: Optimal Contract Design for Collaborative Machine Learning," AAAI-25 Conference, Philadelphia, USA, Mar 2025.

Teaching

Teaching Assistant, Department of Mathematics, Columbia University, New York, NY, 2017–2018.

Undergraduate Tutor, Department of Statistics, Columbia University, New York, NY, 2017–2018.

Program Coordinator, Wall Street Training Program, GEC Academy, Beijing, China, Jul–Aug 2017.

Public Writing

South China Morning Post

"Hong Kong's Covid-19 testing regime must be refined so that it's not all stick, no carrots," Feb 4, 2021.

"Why death of George Floyd should make the world take a good look at itself," Jun 14, 2020.

"For Hong Kong, the only way out of a prisoner's dilemma is to give and take," Nov 29, 2019.

Awards and Honors

Joseph and Norma Preziosi Endowed Scholarship, Columbia University, 2018
Dean's Scholarship, College of Science and Engineering, City University of Hong Kong, 2017
HKSAR Government Scholarship, City University of Hong Kong, 2015–16
Chan Feng Men-ling Chan Shuk-lin ELC Scholarship, City University of Hong Kong, 2014
Full Tuition Scholarship, City University of Hong Kong, 2014, 15, 17

Technical Skills

Programming: Python, R, MATLAB, Stata, OxMetrics, Git; pandas, NumPy, scikit-learn, PyTorch

Tools: LaTeX/Overleaf, Notion, VS Code, Google Cloud, Alibaba Cloud, LLMs (coding/debugging)

Software Development: CLI design, YAML configuration, JSON/JSONL artifacts, schema validation, streaming data ingestion, parallel processing, API-based experimentation, token/cost tracking, automated reporting, Markdown/Sphinx documentation

Languages: English, Mandarin Chinese, Cantonese (intermediate)